

SOURCE INVENTORY

Enterprise Manufacturing Data Platform (EMDP)

1. Document Control

Document	Source Inventory
Project	Enterprise Manufacturing Data Platform (EMDP)
Sprint	Sprint 1 – Data Discovery & Ingestion
Purpose	Identify, classify, and document the source datasets used for Sprint 1 ingestion.

2. Inventory Overview

The Source Inventory records the datasets selected for the EMDP Sprint 1 ingestion activities. The source set covers the five required ingestion formats—CSV, Excel, JSON, XML, and SQL—and represents manufacturing, IoT, sales, ERP purchase-order, and ERP inventory information.

3. Source Inventory

Source ID	Source / File	Format	Records	Fields	Business Area	Ingestion Method	Status
SRC-001	iot_manufacturing_data.csv	CSV	100,000	13	IoT / Manufacturing	Pentaho CSV ingestion	Ready
SRC-002	manufacturing_dataset.xlsx	Excel	3,000	20	Production / Quality	Pentaho Excel ingestion	Ready
SRC-003	sales_pipeline.csv	CSV	8,800	8	Sales / Enterprise	Pentaho CSV ingestion	Ready
SRC-004	sales_pipeline.json	JSON	8,800	8	Sales / Enterprise	Pentaho JSON ingestion	Ready
SRC-005	erp_purchase_orders.xml	XML	1,000	11	ERP / Procurement	Pentaho XML ingestion	Ready
SRC-006	erp_inventory_source.sql	SQL	1,000	10	ERP / Inventory	SQL source → Pentaho Table Input	Ready

4. Source Descriptions

SRC-001 – IoT Manufacturing Data

File: `iot_manufacturing_data.csv`

Format: CSV

Volume: 100,000 records, 13 fields

Purpose: Represents machine and IoT telemetry used for manufacturing monitoring, including machine status, sensor readings, anomaly indicators, remaining-life information, failure information, downtime risk, and maintenance indicators.

SRC-002 – Manufacturing Dataset

File: `manufacturing_dataset.xlsx`

Format: Excel

Volume: 3,000 records, 20 fields

Purpose: Represents manufacturing production information, including production records, product type, machine, shift, output, defects, production time, cost, energy, downtime, maintenance, quality checks, temperature, and humidity.

SRC-003 – Sales Pipeline CSV

File: `sales_pipeline.csv`

Format: CSV

Volume: 8,800 records, 8 fields

Purpose: Represents sales opportunity information including opportunity, sales agent, product, account, deal stage, engagement date, close date, and close value.

SRC-004 – Sales Pipeline JSON

File: `sales_pipeline.json`

Format: JSON

Volume: 8,800 records, 8 business fields per record

Purpose: Provides the sales-pipeline data in JSON format and is used to demonstrate JSON ingestion. It represents the same logical sales-pipeline dataset as SRC-003 in a different physical format.

SRC-005 – ERP Purchase Orders

File: `erp_purchase_orders.xml`

Format: XML

Volume: 1,000 records, 11 fields

Purpose: Representative/sample ERP procurement source containing purchase-order, supplier, material, quantity, pricing, date, warehouse, and status information.

SRC-006 – ERP Inventory

File: `erp_inventory_source.sql`

Format: SQL

Volume: 1,000 records, 10 fields

Purpose: Representative/sample ERP inventory source containing material, warehouse, stock quantity, reorder level, cost, update date, and stock status information.

5. Source-to-Project Mapping

Project Source Category	Mapped Source	Remarks
IoT Sensor Gateway	SRC-001	Directly represents machine/IoT telemetry.
Production / Manufacturing	SRC-002	Directly represents production and manufacturing records.
Quality-related data	SRC-002	Contains defects and quality-check fields; no separate QMS extract is currently included.
Sales / Enterprise Data	SRC-003 and SRC-004	Same logical sales pipeline represented in CSV and JSON.
ERP / Procurement	SRC-005	Representative XML purchase-order source.
ERP / Inventory	SRC-006	Representative SQL inventory source.

6. Repository Placement

The source files should be retained under the project's datasets/ directory:

- datasets/iot_manufacturing_data.csv
- datasets/manufacturing_dataset.xlsx
- datasets/sales_pipeline.csv
- datasets/sales_pipeline.json
- datasets/erp_purchase_orders_1000.xml
- datasets/erp_inventory_source.sql

7. Ingestion Notes

- Raw source files should remain unchanged in the datasets/ directory.
- Each source will have a dedicated Pentaho ingestion transformation or appropriate input step.
- The CSV, Excel, JSON, XML, and SQL sources will be loaded into PostgreSQL Bronze/staging structures.
- The sales pipeline CSV and JSON represent the same logical business dataset in different physical formats; they should not be counted as two independent business datasets.

- The XML and SQL sources are representative/sample sources created for the project to satisfy the required ingestion-format coverage and will be documented transparently as such.

8. Initial Data Discovery Observations

- The IoT manufacturing CSV contains 100,000 records and provides substantial volume for ingestion and later profiling.
- The manufacturing Excel dataset contains 3,000 records and includes production, defect, downtime, maintenance, and quality-related fields.
- The sales pipeline CSV and JSON each contain 8,800 records and provide a useful demonstration of CSV and JSON ingestion using the same logical dataset.
- The XML purchase-order source contains 1,000 representative ERP procurement records.
- The SQL inventory source contains 1,000 representative ERP inventory records.
- Detailed column definitions, data types, keys, and business meanings will be documented in the Data Dictionary.

10. S1-1 Completion

S1-1 – Analyze all source systems is considered complete for the currently selected source set. The inventory identifies the source files, formats, volumes, business areas, ingestion methods, and source-to-project mappings required to begin Sprint 1 pipeline development.